

[6584]-54

B.E. Civil

QUANTITY SURVEYING CONTRACTS AND TENDERS **(2019 Pattern) (Semester - VIII) (401012)**

Time :3 Hours]

[Max. Marks : 70

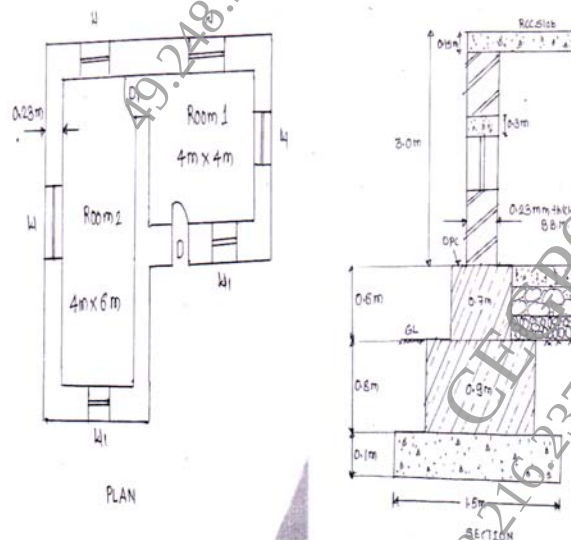
Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to Right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Explain different methods of taking out quantities. Explain any one with example. **[6]**

b) Figure shows plan and section of residential building, Determine the quantities of following item By PWD Method (Assume any suitable data) **[12]**

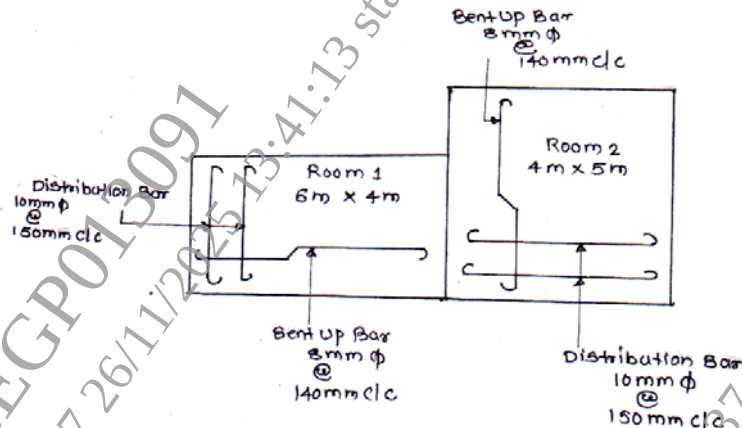
- i) Excavation in foundation
- ii) PCC in Foundation
- iii) UCR masonry in CM (1 : 6) in foundation
- iv) CR masonry in CM (1 : 6) in foundation
- v) Damp proof course 3 cm thick



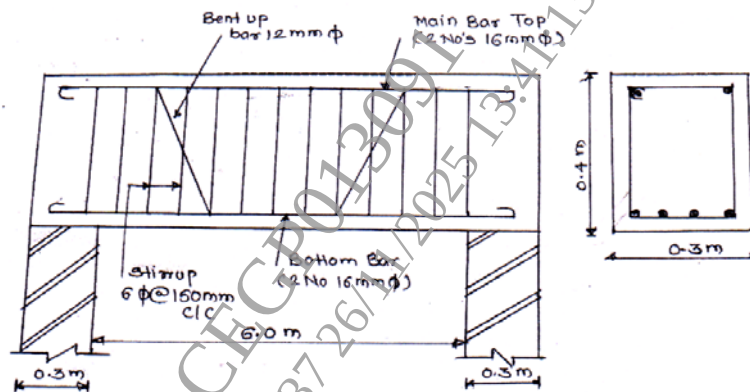
P.T.O.

OR

- Q2) a) Determine quantity to steel reinforcement in slab for both room from figure clear cover on both side 20 mm (Assume any suitable data) [9]



- b) R.C.C simply supported beam of side 300 mm × 400 mm is shown in figure The span of beam is 6.0 m and end bearing (both sides) is 30 cm. calculate total quantity of steel reinforcement. (Assume any suitable data) [9]



- Q3) a) Differentiate between following methods : [8]

- Mid sectional area method & Mean Sectional area method.
- Trapezoidal formula method & Prismoidal formula method.

- b) Calculate the quantities of Earthwork is cutting and filling for road with following data : [10]

- Formation width of road is 10 m
- Slope in cutting 1 : 2
- Slope in Banking 2 : 1

Chainage	0	30	60	90	120
GL in m	400	398	50.50	396	392
FL in m	397	396	395	394	393

OR

- Q4) a)** List out different methods to workout Earth work quantity for road work. Explain any one method. [8]
- b)** Prepare an estimate of a pipe culvert for following item as shown in Fig. 4 (a) and Fig. 4(b) [10]
- Excavation
 - Earth filling
 - Rubble soling

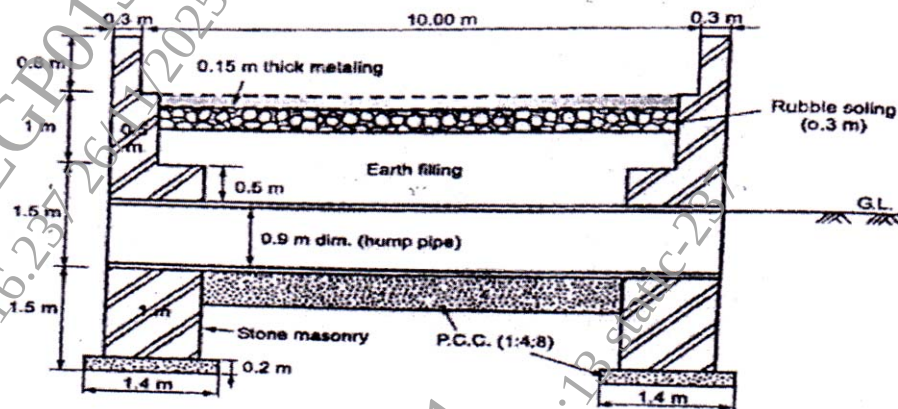


Fig 4(a)

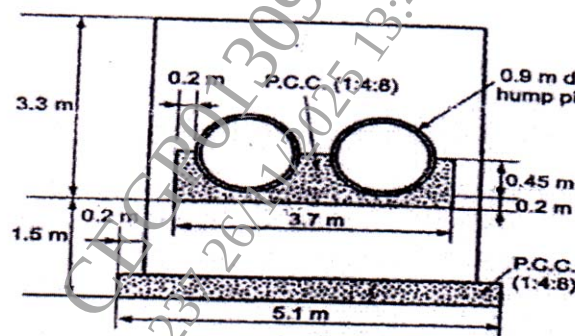


Fig 4(b)

- Q5) a)** Prepare a rate analysis of 12mm thick cement plaster (1:4) [9]
- b)** What is the necessity of drafting specification for Civil engineering work. Explain [8]
- General specification.
 - Detailed specification.

OR

- Q6) a)** Prepare a rate analysis of Cement concrete 1:2:4 for R. C. C. Roof Slab with 2.5 % steel. [9]
- b)** Prepare a detailed Specification for plain cement concrete & Brick Masonry in Superstructure. [8]

- Q7) a)** Why do depreciation occur in the valuation of property? What are the different methods of calculating depreciation. Explain any one method of calculating depreciation stating formula used, merits & demerits. [6]
- b)** Explain with example five purpose of valuation. [5]
- c)** A building is constructed at a cost of 10 lakhs. The life of building may be assumed to be 90 years and the scrap value of building to be 10% of building cost. Determine the depreciation in 60th year. Use straight line method, sinking fund method assuming 10% compound interest. [6]

OR

- Q8) a)** Differentiate between Cost, Price & Value with suitable example. [6]
- b)** Define the following terms : [5]
- i) Dual rate Year purchase
 - ii) Sinking Fund
 - iii) Annuity
 - iv) Obsolescence
 - v) Free hold Propert
- c)** The depreciated replacement value of building has to be found out with the following data : [6]
- i) Total builtup area = 1500 sq-m
 - ii) Age of building = 50 years
 - iii) Life of building = 100 years
 - iv) Scrap value after useful life = 10%
 - v) Percentage for sinking fund = 8%
- Assume rate of construction as Rs. 4000/sq-m
